# Paper A-14: Earth Volcanology: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD
Independent Researcher
ORCID: 0009-0001-0556-5516

May 2026

**Abstract**

This paper asks whether Constraint-Driven Flux Dynamics (CDFD) and Adaptive Flux Limitation (AFL) can give the named system a sharper audit language. The working variables are driving flux $\Phi$, constraint $C$, surface responsiveness $S$, and structural memory $M_s$. The useful question is plain: what can be measured, what would count as overload, and what result would make the mapping fail?

## 1 Series Context

Part I sets the flow–constraint language used here [1]. Part II develops the Life Number and the origins-of-life use of the same notation [2]. Part III applies AFL to biology and medicine [3]. Part IV is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

## 2 Introduction

**Notation.** In Part IV, $C$ names the active constraint or capacity burden in the system being discussed. $\Phi$ is the driving flow, $S$ is the surface response, and $M_s$ is retained structural memory. $\Lambda$ appears only where the Part II Life Number is being referenced [2]. Other Greek symbols are local coefficients whose meaning is set by the equation in which they appear.

### 2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [1], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux ($\Phi$) against a rigid topological constraint ($C$) can drive system saturation. When $\Phi/C$ approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

### 2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory ($M_s$) and its relaxation parameter decays ($\tau_{relax} \to \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Magma generation deep in the Earth's mantle creates buoyant, low-density melts that ascend toward the surface. During this ascent, pressure decreases, allowing dissolved volatiles (such as $H_2O$ and $CO_2$) to exsolve and form bubbles. The dynamics of these bubbles, and the structural integrity of the surrounding rock, dictate whether a volcano will gently erupt lava or explosively detonate.

Applying CDFD to volcanology reveals that a volcano is functionally a high-pressure nozzle attempting to regulate thermodynamic throughput.

## 3 The Magmatic Conduit as an Adaptive Surface

We define the thermodynamic state of the magma chamber and conduit using the Universal Adaptive Ratio:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \tag{1}$$

In volcanic systems:

- **Flux ($\Phi$)**: The upward flow of magma and, critically, the volumetric expansion flux of exsolving gases.
- **Constraint ($C$)**: Lithostatic confining pressure, magma viscosity, and the structural rigidity of the vent caprock.
- **Responsiveness ($S$)**: The elasticity of the magma chamber walls and the ability of the magma to deform (low viscosity).
- **Memory ($M_s$)**: Crystallization and cooling. As magma stalls, it cools and crystallizes, increasing its viscosity and forming a solid plug. This physical state locks the system's memory, severely cementing the constraint.

### 3.1 Effusive vs. Explosive Eruptions

In a basaltic shield volcano (like those in Hawaii), the magma has low viscosity. The constraint $C$ remains low, and the system can continuously process the upward flux $\Phi$ smoothly. The system remains near $\Psi_s \approx 1$. The eruption is effusive and constant.

Conversely, in a stratovolcano with highly viscous, silica-rich magma, the system behaves pathologically. As gases exsolve ($\Phi$ spikes), the high viscosity prevents the bubbles from escaping. The magma cools, crystallizing into a solid plug at the vent (high $M_s$). The constraint $C$ becomes absolute.

The system enters a chronic choke state ($\Psi_s \ll 1$). The upward flux from the mantle continues to compress into the chamber. Because the constraint $C$ cannot flex, the internal pressure exceeds the fracture strength of the rock. The constraint fails catastrophically, dropping $C$ to zero in an instant. The pressurized gas expands supersonically, resulting in a Plinian explosive eruption that atomizes the magma into ash.

# 4 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. $\Phi$ may be a rate of material, energy, information, capital, or signal flow. $C$ is the bottleneck that slows or redirects that flow. $S$ measures the system's ability to reconfigure under stress, and $M_s$ records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

## 4.1 Current Runtime Diagnostic: Universal Network Cascade

The diagnostic script `Part_E_Synthesis/supplementary/run_partiv_discovery.py` keeps this paper tied to the current CDFD Runtime [4], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the May 2026 run, the localized hub-drive stress test ended with hub $\Psi_s = 14.8$, peak network $\Psi_s = 14.8$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

# 5 Conclusion

Volcanic eruptions are thermodynamic venting mechanisms. By applying the CDFD equation, we see that effusive eruptions represent a healthy, regulated flow regime, while explosive eruptions represent the catastrophic failure of an over-constrained system locked by magmatic memory. The Earth regulates its internal heat flux using the exact same topological mathematics that a biological cell uses to regulate its metabolic gradients.

# Conflict of Interest

The author declares no conflict of interest.

# Data Availability

Release-local runtime diagnostics are available under each Part's `outputs/` folder, with release-wide synthesis artifacts under `Part_E_Synthesis/supplementary/`, `Part_E_Synthesis/outputs/`, and `Part_E_Synthesis/figures/`.

# 6 Reference Layer

## References

[1] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026.

[2] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026.

[3] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026.

[4] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026.